

Highway Engineering

A) Classification of Roads

1) Urban Roads = A road within a City or town

a) Express ways (or) Motor way:

Central portion of a road for high speed vehicles

→ Parking, loading & unloading of goods are not permitted

→ Pedestrian crossing are also not allowed

b) Arterial Streets:

A Continuous Major road in a city having high traffic

(eg: Bander road, Eluru road in Vijayawada)

c) Collector Streets:

Collecting & distributing the traffic to and from local streets.

→ These are located in residential, industrial & commercial areas

→ No restrictions for parking & loading

2) Non-Urban Roads = A road outside the City or town

a) National Highways:

→ These are Major & important roads of a Country

→ Connecting important places (like State Capital, airports... etc)

→ Organised by CPWD (Central Government)

b) State Highways

→ Connecting district headquarter & neighbouring states

→ Organised by PWD (State Government)

c) Major district road & d) Village Road

* Organisations:

1. First major/longest road = Kolkata - Lahore

2. IRTDA = Mumbai (Head office), Under Jayakar Committee

3. CRO in 1934

4. IRC in 1943, New Delhi (Head office)

* Highway Planning in India

1. Nagpur Road Plan (or) First 20-year Road Plan (1943-63)

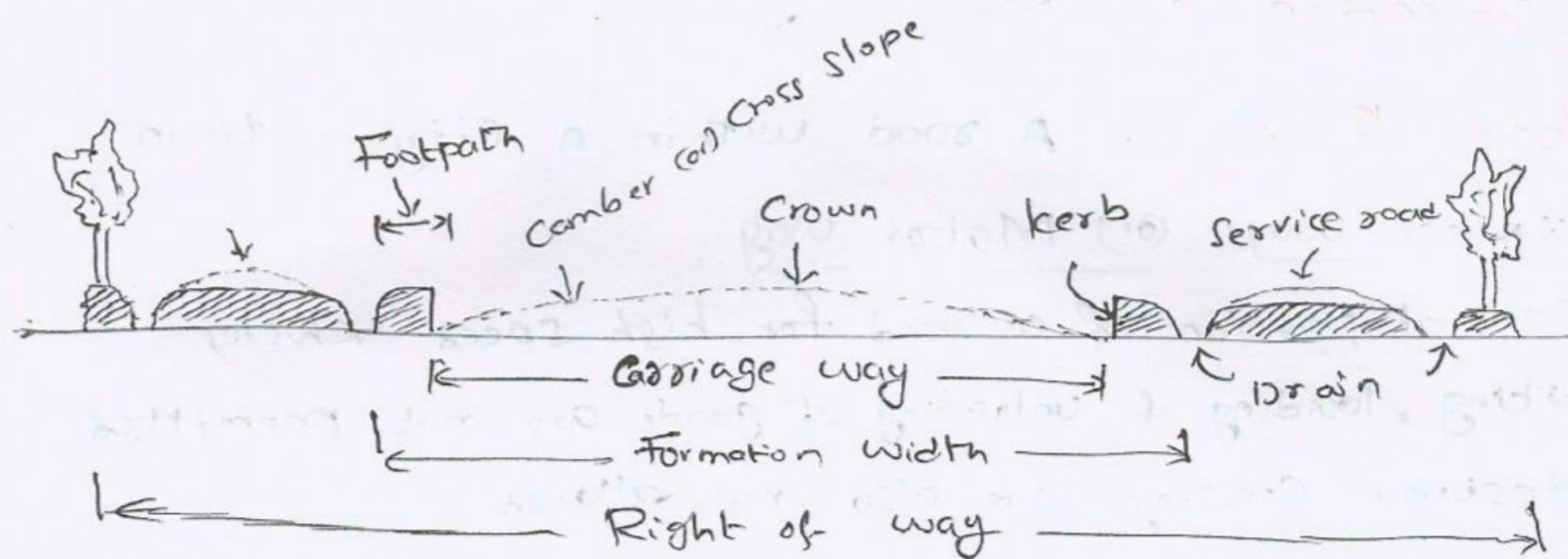
→ Recommended "Star & Grid" formulae

→ Road length achieved by 1961

2. Bombay Plan (or) Second 20-year Road Plan (1961-81)

→ 5 different formulae for diff roads

#1 Components of a City Road



1. Right of way

The area of land acquired for construction & future development of a road.

2. Formation width

Bottom width of highway cutting excluding side drains.
i.e. Formation width = Carriage way + Shoulders

3. Carriage way (or) pavement:

Portion on road surface used for vehicular traffic

For single lane road = 3.75m

Single lane Bridge road = 4.25m

4. Shoulders

Between outer edge of Carriage way - inner edge of side drains

→ Footpaths are the portions in shoulders

→ prevent edges of the road from wear & tear

→ Minimum Shoulder width = 2.5m (IRC)

5. Crown

Highest point on the road surface

6. Kerbs:

Rises portion/boundary between carriage way & shoulder

→ Also provides between pavement & separators/dividers.

7. Side Slopes

= 1:1.5



Camber: or Cross-Slope or Cross Fall

The Slope of line joining Crown & Edge of the road Surface.
 → To drain off the water from road surface (main object)

Sloped Camber = 

Two straight slopes joining center (For CC pavements)

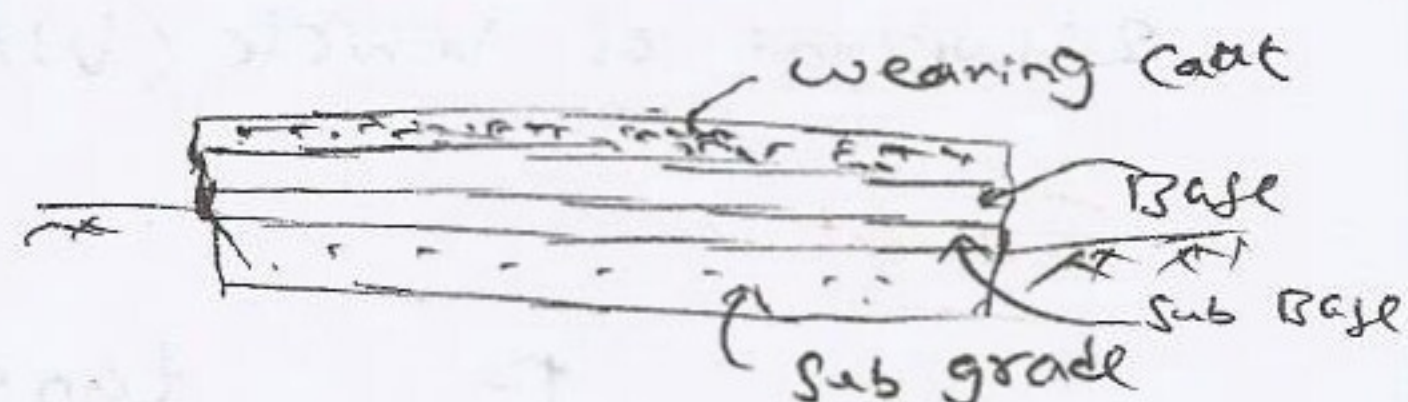
Barrel Camber = 

Continuous curve either parabolic/elliptical

IRC Recommendations

Type of Road Surface (Permeability factor)	Range of Camber	
	Heavy rainfall	Light rainfall
1. CC & high bitumen	1 in 50 (2%)	1 in 60 (1.7%)
2. Thin bitumen	1 in 40 (2.5%)	1 in 50 (2%)
3. WBM & gravel pavement	1 in 33 (3%)	1 in 40 (2.5%)
4. Earth Surface	1 in 25 (4%)	1 in 33 (3%)

Components of Road Surface



1) Sub Grade:

- The ground on which foundation of a road rests.
- It is provided by digging up the sub-soil
- The strength & durability of ~~soil~~ road depends upon its sub-grade
- Entire load ultimately transferred to sub-grade

2) Sub-Base:

- Additional layer b/w sub-grade & Base
- provided when Bearing capacity of soil is poor & high traffic

3) Base or Soiling (or) Foundation of Road:

- It receives impact of traffic through wearing course
- Base course ≥ 30 cm and ≤ 10 cm (for flexible)
- Rigid (in case of rigid pavement) & Broken stone (in case of flexible pavement)

4) Wearing Course

- The Superstructure of the road
- Wearing course ≥ 10 cm
- It should be impervious, friction, abrasion & resistance to impact of wheels.

Super Elevation

Super Elevation (or) Cant or Banking

When a vehicle negotiates a curved path, it is subjected to a centrifugal force. To resist this outer edge of curve is raised above the inner edge.

Forward tilt given to the total length of curve

- In the absence of Super elevation pot holes are put across the outer edge of the road curve.
- Super elevation $\neq$ Camber
- Greater super elevation inconvenience to slow moving traffic.
- Max super elevation = 1 in 15.

Forces acting on vehicle while moving in curve

1. Centrifugal force ($P = \frac{Wv^2}{gR}$)

2. weight of vehicle (W) , 3. Frictional force

Equation:

$$\frac{P}{W} = \frac{\tan \theta + f}{1 - f \tan \theta}$$

$$\Rightarrow \frac{P}{W} \approx \tan \theta + f = e + f$$

$$\therefore e + f = \frac{v^2}{gR}$$

(or)

$$e + f = \frac{v^2}{127R}$$

($\because v$ is in kmph)

where

$e = \tan \theta =$ Super elevation

$f =$ Lateral frictional Coefficient = 0.15

$v =$ Speed of vehicle (m/s)

$R =$ Radius of curve (m)

$$\therefore e = \tan \theta = \frac{\text{Rise}}{\text{Width of pavement}}$$

Equilibrium Super elevation ($\because$ Neglect friction, i.e. $f=0$)

$$\therefore e = \frac{v^2}{127R}$$

Maximum Super elevation (IRC)

According to IRC Super elevation balances centrifugal force corresponding to maximum of 75% (i.e. $\frac{3}{4}$ th) of the design speed. $\Rightarrow e = \frac{0.75v^2}{127R}$

$$\therefore e = \frac{v^2}{225R}$$

Gradients

Gradient of the road depends upon

- Rainfall of the locality
- Nature of traffic
- Nature of ground.

1. Minimum gradient

→ For the purpose of easy drain off rain water

Minimum gradient for Flexible pavement (1 in 250)

Rigid pavement (1 in 300)

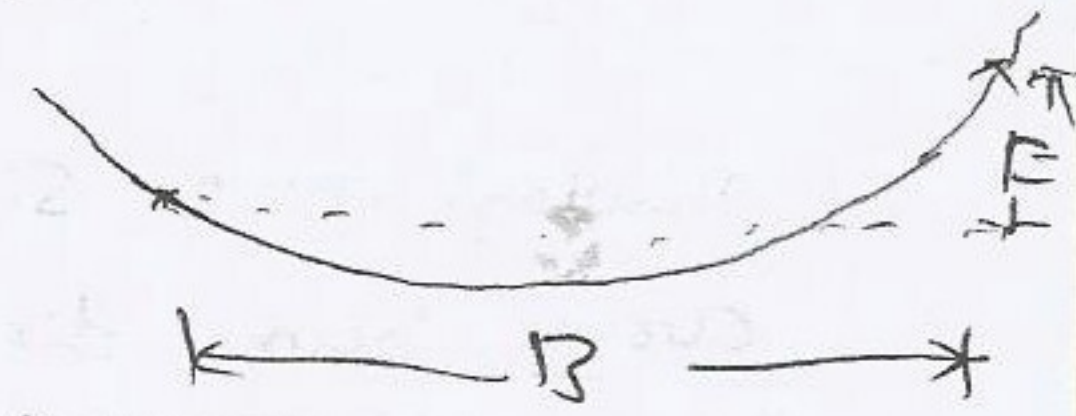
2. Maximum or Limiting gradient

Max/Steepst gradient allowed on road, when

- To avoid deep excavation
- To avoid long detours

3. Average gradient

$$= \frac{\text{Rise/fall b/w any two points in the alignment}}{\text{Horizontal distance b/w two points}} = \frac{E}{B}$$



4. Exceptional gradient

The gradient $<$ Minimum gradient &
 $>$ Maximum gradient

→ Necessary to avoid deep cuttings

→ Max length of Exception gradient $\neq$ 60m per one km.

5. Ruling gradient

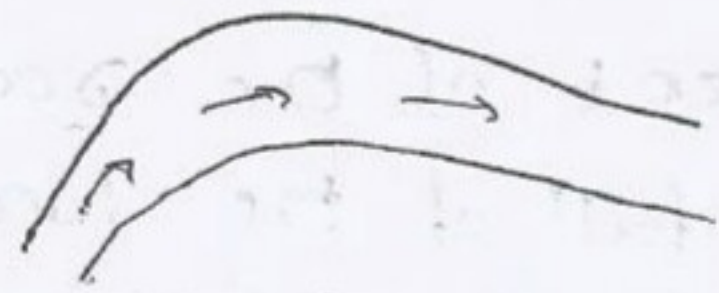
The engineer designed gradient after all factors considered

IRC Recommendations

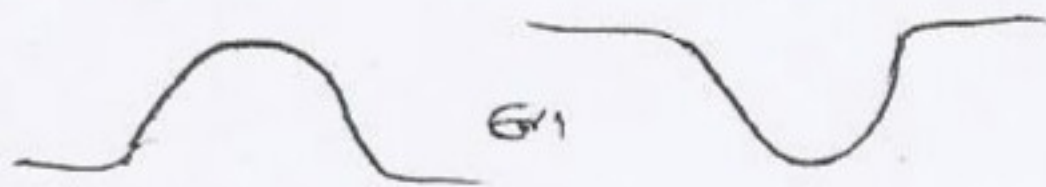
Nature of Terrain	Ruling gradient	Limiting gradient	Exceptional gradient
1. Plains	1 in 30	1 in 20	1 in 15
2. Hills	1 in 20	1 in 17	1 in 14
3. Steep terrain	1 in 17	1 in 14	1 in 12.5

Curves

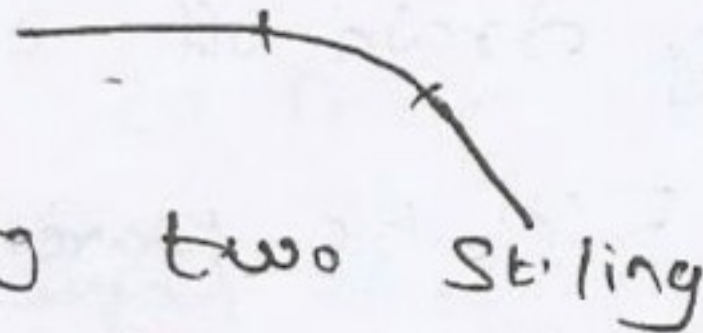
1. Horizontal Curve or Transition Curve
Change in direction



2. Vertical Curve
Change in gradient



1) Simple Curve
Single arc connecting two straight



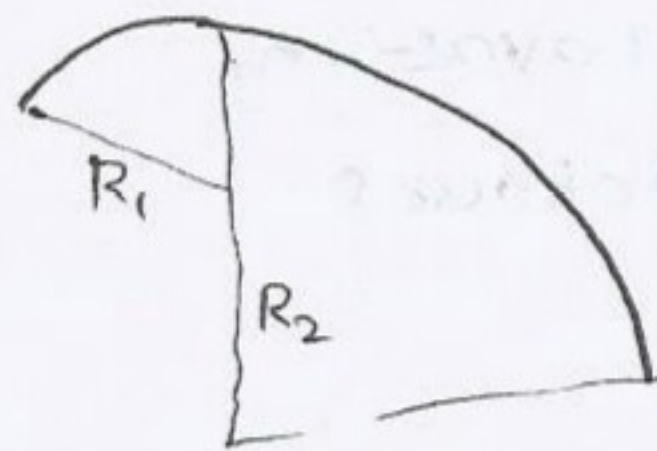
2. Reverse Curve



Two simple curves opposite in direction joint at common tangent point

3. Compound Curve

Two (or) more simple curves same direction (with diff Radius) joining at common tangent point.



1. Transition Curve:

Radius gradually changes from infinite value to finite value or vice versa for giving easy change of direction is called a transition curve or spiral or easement curve.

$$\text{Length of Transition Curve } (L) = \frac{\sqrt{3}}{48CR}$$

$$\text{Rate of change of acceleration } (C) = \frac{80}{70+V} \in [0.5, 0.8] \text{ (m/s}^3\text{)}$$

a) True Spiral or Clothoid

It is an ideal shape of transition curve

b) Cubic Spiral

Transition curve recommended by IRC is "Cubic Spiral"

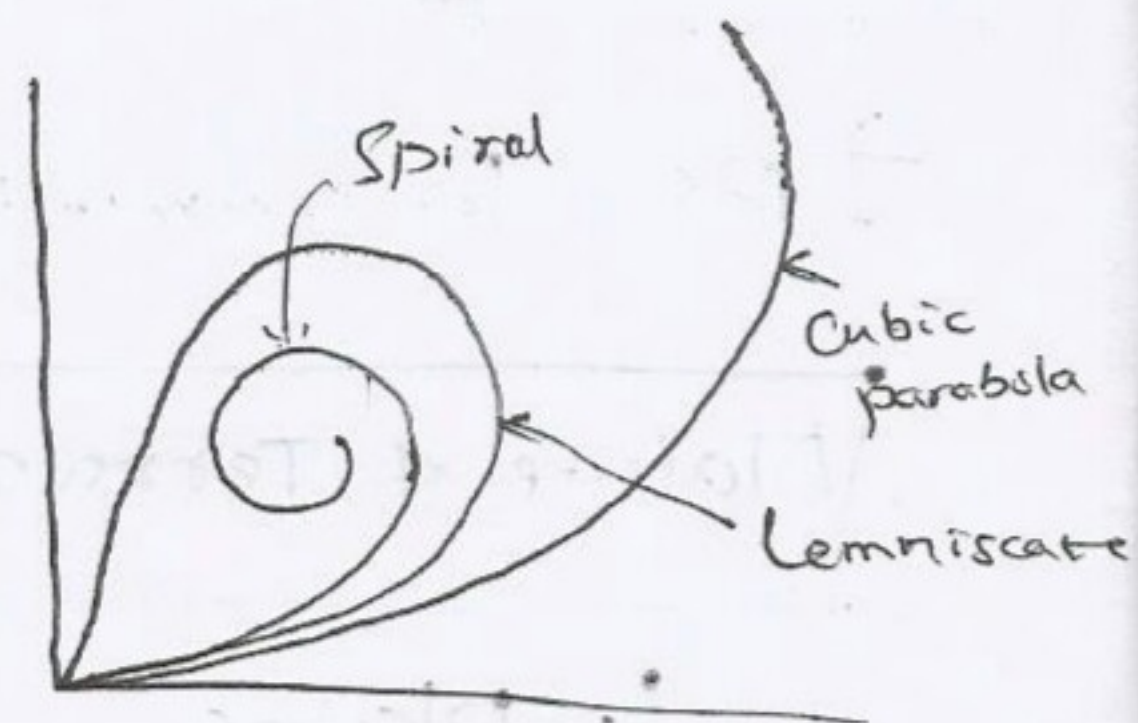
$$\text{Polar deflection angle } (\alpha) = \frac{1}{3} \text{rd of Deviation angle of the curve } (\phi)$$

c) Cubic parabola or Froude's Transition/easement curve

d) Bernoulli's Lemniscate curve

Most used in modern road construction & when deflection angle b/w tangents is large

$$\text{Polar equation } (r) = \sqrt{2RL \sin 2\alpha}$$



2. Vertical Curves

Due to change in grade in the vertical alignment of highway

a) Summit (or) Crest curves

Convexity upwards

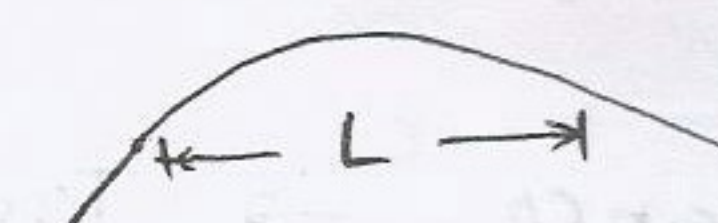
Required to introduce at

→ +ve grade meets -ve grade

→ +ve meets middle +ve

→ +ve meets level stretch

→ -ve meets steepen -ve



$$N = n_1 + n_2$$



$$N = n_1 - n_2$$



$$N = n_1$$



$$N = n_2 - n_1$$

$$\text{Safe OSD } (L) = \frac{GS^2}{9.6} \quad (S < L)$$

$$= 2S - \frac{9.6}{G} \quad (S > L)$$

($\therefore S = \text{Sight Distance}$)

$$\text{Safe SSD } (L) = \frac{GS^2}{4} \quad (S < L)$$

$$= 2S - \frac{4}{G} \quad (S > L)$$

b) Valley (or) Sag curves

Required to introduce @

→ -ve grade meets +ve

→ -ve meets middle -ve

→ -ve meets level

→ +ve meets steepen +ve



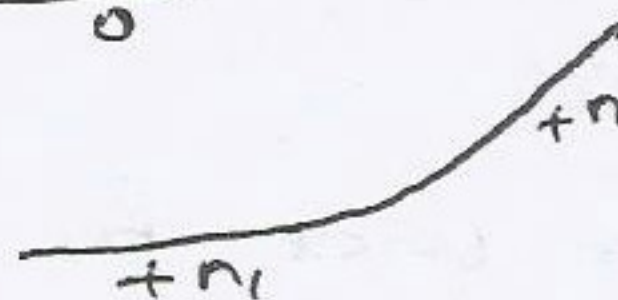
$$N = -(n_1 + n_2)$$



$$N = -n_1 - (-n_2) = n_2 - n_1$$



$$N = -n_1$$



$$N = n_1 - n_2$$

According to IRC

Based on Comfort conditions

$$\text{Length of valley curve } (L) = 2 \times \text{length of transition curve } (L_s)$$

SSD = head light beam distance

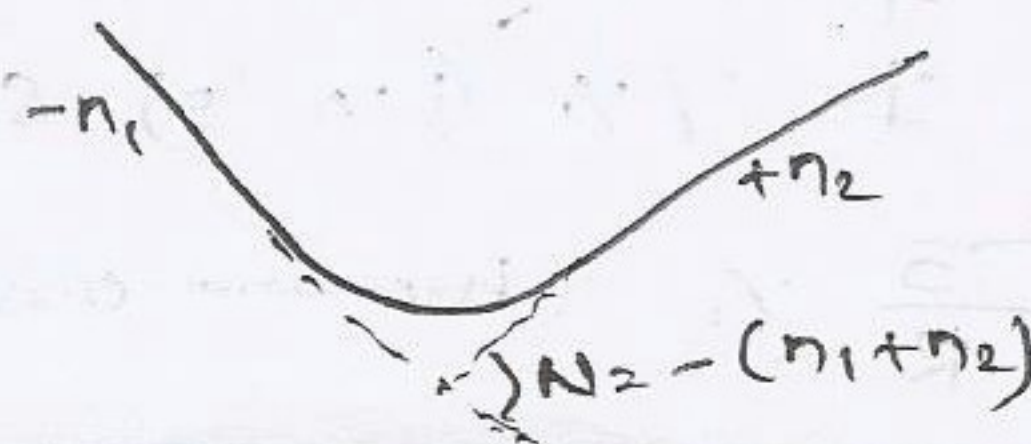
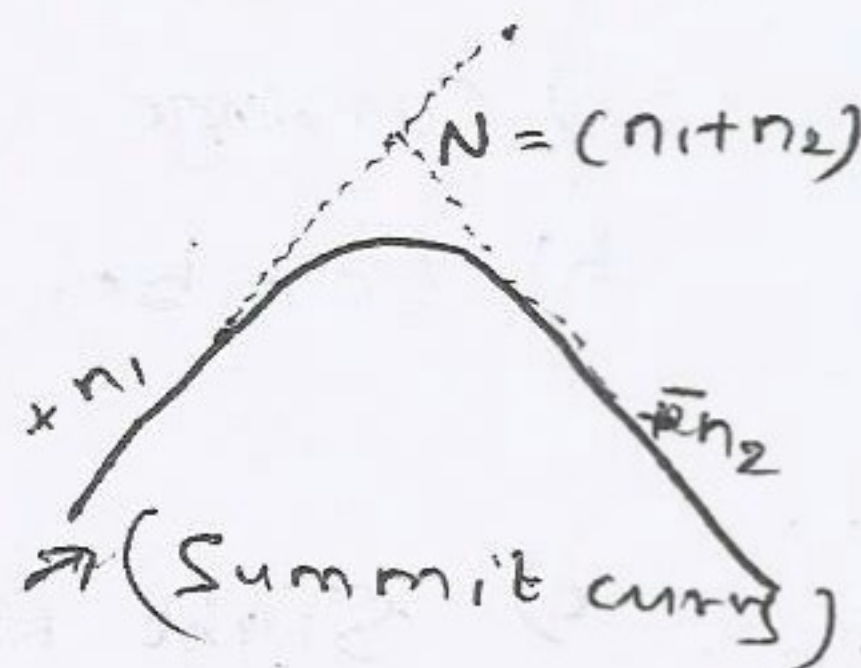
Head light = 0.75 m (from Road Surface)

3. Deviation angle [i.e. $N = n_1 - n_2$]

Deviation angle maximum when

a) Ascending gradient meets descending gradient (Summit curve)

b) Descending gradient meets ascending gradient



Valley curve

Hill Roads

1. Fixing alignment of Hill Road

Reconnaissance $\rightarrow$ Trace Cut $\rightarrow$ Detailed Survey
(Paper work) (Prepare inspection road = 1-1.2m) (Field Survey)

2. Terms used in Hill Road

- a) level terrain cross-slope $< 10\%$.
- b) Rolling terrain cross-slope $10-25\%$.
- c) ^{Mountainous} ~~Steep~~ terrain cross-slope $25-60\%$.
- d) Steep terrain cross-slope $> 60\%$.

e) Salient Curve Convex Shape () ()

f) Re-entrant Curve Concave Shape ()

g) Floating gradient

The gradient at which no tractive effort is required to maintain constant speed (i.e. slightly -ve gradient)

Gradient $>$ Floating gradient (need to increase speed)
 $\hookrightarrow$ maintain constant speed

Gradient $<$ Floating gradient (need tractive effort (brake))
 $\hookrightarrow$ maintain constant speed

3. Classification of Hill Road (Boarder road organisation)

- a) National Highways
- b) Class 9 (6m wide for 3-tonne vehicle)
- c) Class 5 (4.9m " " 1-tonne ")
- d) Class 3 (2.45-3.65m " for Jeeps)

4. Standards of Hill Road :

- a) Carriage way $\nless 3.75m$ or $\approx 3m$ (for Jeeps)
- b) Camber = Steep Camber (Generally)
 $\approx$ Flatten Camber (when longitudinal slope > 1 in 20)
- c) Sight Distance = SSD
- d) Curve radius $\nless 30m$
- e) Super elevation $\nless 10\%$ (1 in 10)
 $\nless 7\%$ (1 in 15) skew-bound section
- f) Gradient $\nless \frac{75}{R} \%$ (Horizontal curve)

5. Construction of Hill Roads

a) Retaining Walls

Top width = 60 cm

Bottom width = $0.4H$ ($H < 6m$)

= $0.4H + 30cm$ ($H > 6m$)

Batter = 1 in 12 (wall outside)

b) Brest walls

Thickness of wall = 60 cm (Throughout)

c) Parapet walls (valley side)

Thickness of wall = 60 cm (Through out)

Height = 75 cm (From berm level)

Group Index Method

G.I of a soil is a number which reflects the characteristic of the soil.

$$G.I = 0 \text{ to } 20$$

G.I $\propto$ Sub-Base thickness.

California Bearing Ratio (CBR) method

C.B.R of a soil is a percent which gives an idea about quality of material, compared to Excellent material (100%).

CBR apparatus consists of

Mould = 150 mm dia

Cylindrical Plug = 50 mm dia

Penetrate @ 1.25 mm/minute

Value recorded @ 2.5 mm & 5.0 mm.

Concrete pavement

1. Thickness of the slab:

$$t = \sqrt{\frac{3W}{\sigma}} \quad (\text{Older's theory})$$

$$= \sqrt{\frac{2.4 W C}{\sigma}} \quad (\text{Sheet's formula})$$

Where

W = wheel load

σ = Unit Stress in Tension

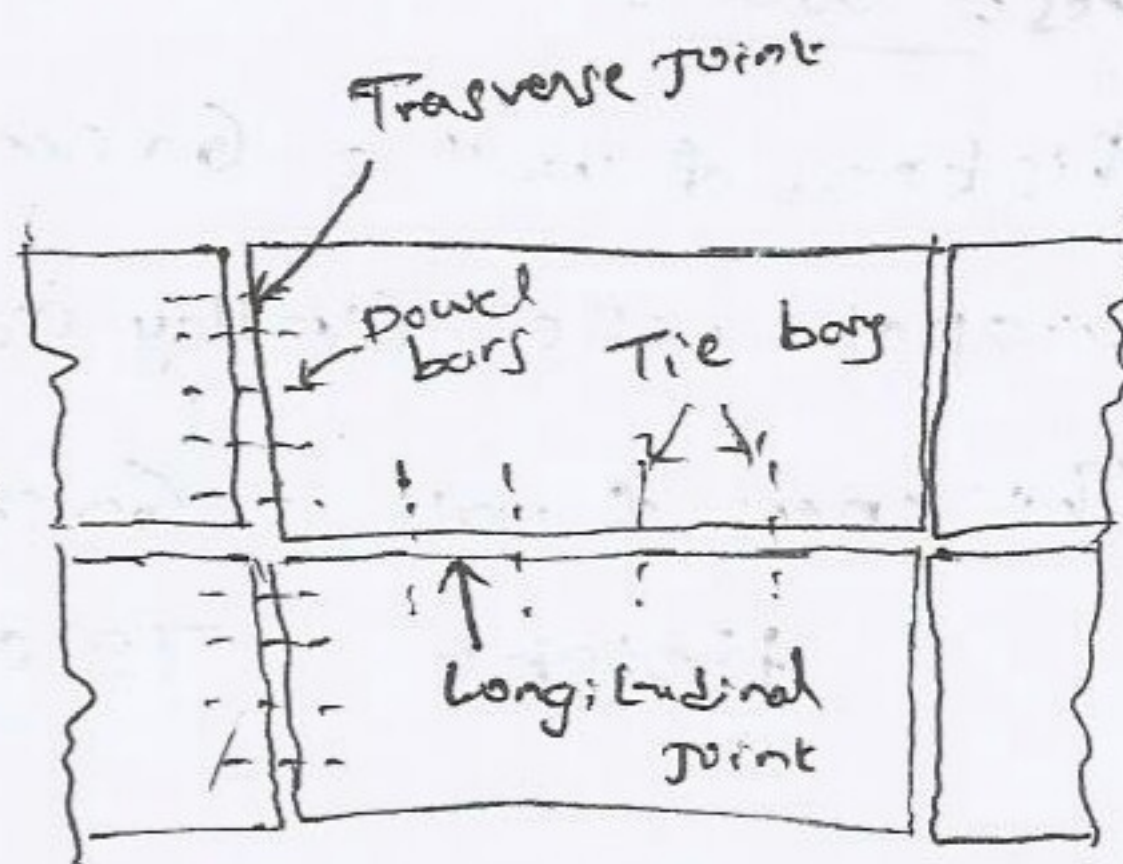
C = Coefficient of Subgrade Support

2. Joint in Concrete pavement

a) Longitudinal joint:

→ Butt joint is simplest longitudinal joint recommended by IRC

→ Tie bars are used to hold the adjacent slab together.



Length of Tie bar $\propto 2 \times$ length req to develop bond strength

$$L_t = \frac{2 a \sigma_s}{2.46 p}$$

($\because a, p$ = Area & perimeter of bar)
 σ_s = allowable working stress

b) Transverse joint

a) Expansion Joint. (induced by Temperature Stress)

These are connected by Dowel bars Spaced 300 mm c/c

b) Contraction joint (induced by moisture changes)

Max Spacing of Contraction joint = 4.5 m (unreinforced cc)

c) Warping or Hinged joints = 14 m (Reinforced cc)

Longitudinal joint with tie bars fall in this class of joint.

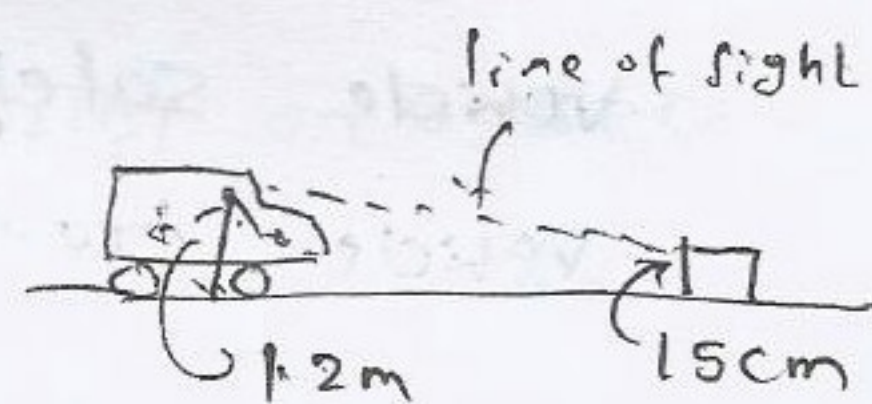
Sight Distance

1. Sight Distance (SD)

The distance, measured along the centre line of a road, over which a driver can see the opposite object on the road.

Line of sight for a driver = 1.2 m

Object can see = 15 cm



2. Stopping Sight distance (SSD)

Stopping or absolute minimum or Non passage or Non-overtaking Distance.

The minimum sight distance required by the driver to stop vehicle after an object on the way becomes visible.

Minimum SSD = Reaction distance + Braking distance

* PIEV Theory:

Total reaction time of driver is split into four parts

- (i) perception (eye & ear - brain)
- (ii) Intellection (Understanding & Comparison... etc)
- (iii) Emotion (fear, anger & ~~decision~~)
- (iv) Volition (Time required to final decision/action)

Total Time reaction = 0.5 Sec (For simple situation)

$$SSD(s) = 0.28 Vt + \frac{0.01 V^2}{\eta}$$

$$= \frac{5}{18} Vt + \frac{V^2}{254 \eta}$$

$$= vt + \frac{v^2}{2g\eta}$$

3. Intermediate Sight Distance (ISD)

Is provided to give limited overtaking opportunity to fast moving vehicles (when OSD not provided)

$$ISD = 2 \times SSD$$

Recommended by IRC

Design speed (kmph)	20	25	30	40	50	60	65	80	100
SSD (m)	20	25	30	45	60	80	90	120	180

4. Overtaking Sight Distance (OSD)

Minimum OSD or Safe passing sight distance

The minimum distance required for overtaking another vehicle safely & without interfering the speed of an opposite vehicle travelling @ design speed.

$$\begin{aligned} \text{OSD} &= d_1 + (\text{reaction distance}) \\ &\quad + d_2 \text{ (Actual overtaking distance)} \\ &\quad + d_3 \text{ (opposite vehicle travelling during } d_2) \end{aligned}$$

or)

$$\begin{aligned} \text{OSD} &= d_1 + d_2 \text{ (one way traffic)} \\ &= d_1 + d_2 + d_2 \text{ (Two way traffic)} \end{aligned}$$

→ If you provide overtaking zone with sign posts

$$\text{Minimum length of overtaking zone} = 3 \times \text{OSD}$$

* According to IRC

Time Req for opposite vehicle (T) to travel (d_3)

$$= \frac{2}{3} \times \text{Overtaking manoeuvre (T+t) to travel (} d_1 + d_2 \text{)}$$

The overtaking manoeuvre takes about (T+t)

$$= 9 \text{ to } 14 \text{ sec (Corresponding to 40 kmph - 100 kmph)}$$

Traffic Characteristics

1. Traffic Volume : = Traffic density $\times$ Traffic Speed.

a) Annual average daily traffic (AADT or ADT)

Average number of vehicle per day passing on a section of road during a particular year.

b) Design Hourly volume (DHSV)

used for design purpose

DHSV = Higher volume of (Volume during most of hours) in an year

2. Traffic Type:

a) Current traffic

= Existing traffic + diverted traffic

b) Increase in traffic

= Normal traffic growth (increase in vehicle year to year)

+ Generated traffic (Construction of new facilities)

+ Developed traffic (improvement in adjacent)

3. Traffic Capacity:

a) Basic Capacity

Max. No. of PCU's moving in one hour without gaps.

b) Possible Capacity

Max. No. of PCU's moving in one hour with ~~design~~^{Some} Speed having no restrictions & no obstructions

c) Practical Capacity.

Max. No. of PCU's moving in one hour with design Speed having general restriction & obstructions.

$\therefore$ Basic Capacity $>$ possible Capacity $>$ Practical Capacity.

d) Theoretical Capacity

No. of vehicle passing any point in one hour per lane

$$\therefore \text{Theoretical Capacity } (C) = \frac{1000V}{S} \quad (\text{PCU's/hour})$$

where

V = Design Speed of vehicle (kmph)

S = Centre-centre spacing of moving vehicle

= Reaction distance + Average length of vehicle.

4. Passenger Car Unit (PCU)

In a mixed flow traffic (like cars, buses, rickshaw, carts, etc.) Consider passenger car as the stand vehicle unit (1.0) to convert other vehicle units (by size & speed). And this unit is called "passenger car unit".

Traffic capacity is PCU/hour (or)

PCU/lane/hour

Traffic density in PCU/km

Tentative PCU values suggested by IRC

Vehicle class	PCU
Passenger Car, Auto, Agriculture tractor	1.0
Bike, Cycle	0.5
Rickshaw	1.5
Bus, Truck, lorry, Tractor with trailer	3.0
Horse vehicle	4.0
Small bullock cart & hand cart	6.0
Large bullock cart	8.0

1. Practical Capacity

→ Single lane (3.75m) + earth shoulder (Both direction) PCU/day
1000

→ Single lane (3.75m) + 1.0m hard shoulder - 2500

→ Two lane (7m) + earth shoulder - 10,000

5. Speed of Vehicles

Running Speed = Distance covered by running time

Spot Speed = Speed of vehicle at a particular point/spot

→ Spot Speed Measured by (i) Enoscope (mirror box supported by Tripod)

Simple & cheap, Not much accurate

(ii) Radar Speed meter

Costly but accurate

6. Vehicle characteristic effect.

a) Overall length of vehicle

Effect on turning radii of curve & OSD

b) Overall width of vehicle

Effect on width of pavement, bridge, other drainage structure

b) Overall height of vehicle

Effect on underpassage & tunnel

d) Weight of vehicle

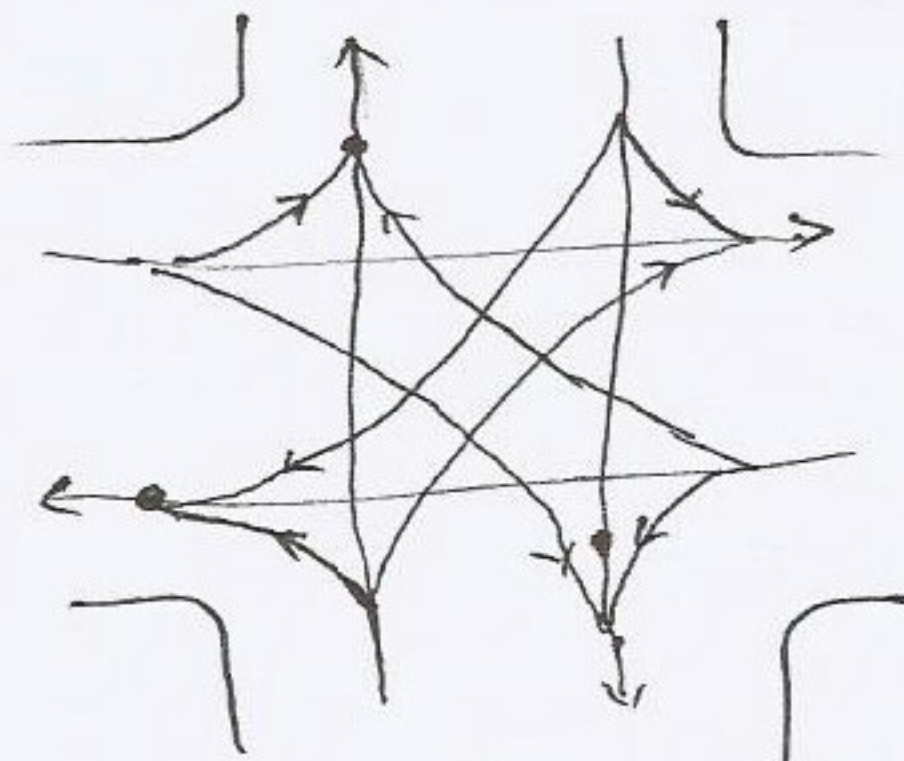
Effect on Thickness of the pavement.

7. Conflicts

No. of intersections

while crossings.

No. of Conflicts



Road A

Road B

Conflicts



4



4



4



12 (3x4)

24

No. of lanes	Both roads Two-way	Both roads one-way
2	24	6
3	24	11
4	44	18

Parking of vehicle is usually preferred at 75° to the aisles.

8. Traffic Signals

a) Signs

- Generally installed at a height of 2.75 m - 2.8 m
- Reverse side of sign plate painted Gray color
- STOP sign has octagonal shape

a) Regulatory (or) Mandatory Signs

Violation of this may cause punishment

("Give way, Restriction and Signs" are also fall in this category)

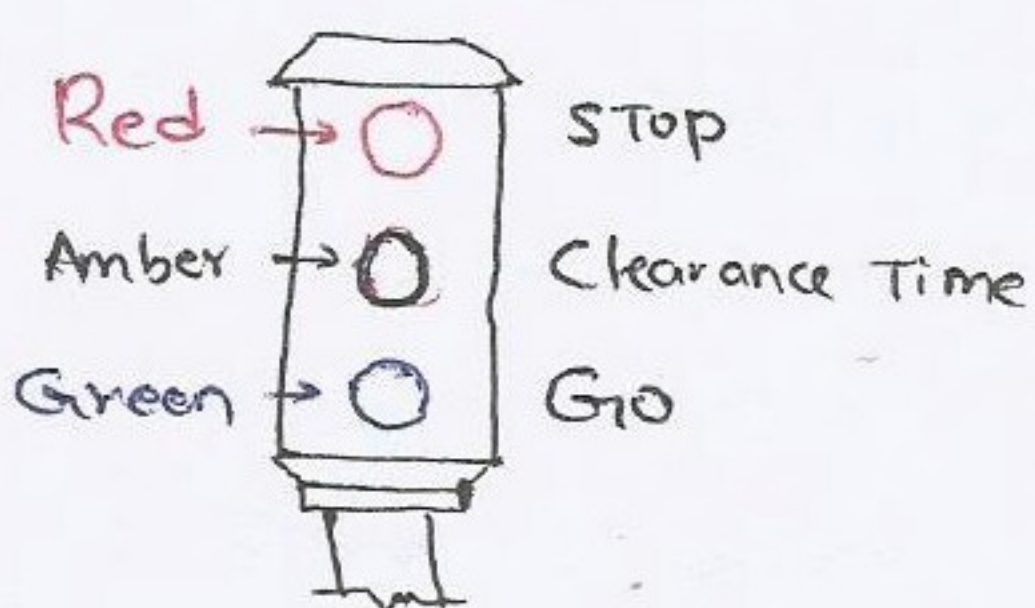
b) Warning Signs

Violation of this may cause accident

c) Informatory Signs

Used to guide the road user. These are Rectangle in shape

b) Signals



Flexible progressive System is the most efficient traffic Signal System.